

A finite-step lattice walk in $\mathbb{Z}^3$ with no three collinear vertices

Erik Kalviainen

August 31, 2026

Abstract

We give an explicit negative answer to Erdős Problem 193. A nested discrete Hilbert path $H: \mathbb{N} \rightarrow \mathbb{N}^2$ carries one of four terminal orientation states. We encode that state twice: as a Gray-code corner appended to the planar coordinates and as a residue modulo four appended to the height. The resulting sequence in $\mathbb{Z}^3$ uses every Hilbert index, has successive steps in one fixed finite set, and has no three collinear vertices. The obstruction is an all-pairs two-adic identity equating a valuation of each planar chord with the ordinary two-adic valuation of its height gap. The complete tagged construction has also been formalized in Lean 4 against Mathlib; the formal theorem uses no additional axioms.

1 Introduction

Let $S \subseteq \mathbb{Z}^3$ be finite. An infinite S -walk is a sequence $(P_a)_{a \geq 0}$ in $\mathbb{Z}^3$ such that $P_{a+1} - P_a \in S$ for every a . Erdős Problem 193 asks whether every such walk must contain three collinear points [1, 2]. Gerver and Ramsey proved the corresponding assertion in dimension two and proved a bounded-collinearity result in dimension three [2].

Our answer to the stated three-dimensional question is negative.

Theorem 1.1 (Main theorem). *There are a finite set $S \subseteq \mathbb{Z}^3$ and an infinite sequence $P: \mathbb{N} \rightarrow \mathbb{Z}^3$ such that*

$$P_{a+1} - P_a \in S \quad (a \in \mathbb{N}),$$

and no three distinct terms of P are collinear.

The construction uses the discrete Hilbert path, but the space-filling property is not the decisive ingredient. What matters is its finite-state quadrant recursion. For indices ending in the same state, a two-adic valuation of the planar chord already equals the valuation of the index gap. We remove the same-state restriction by appending the state itself as one new binary digit in each planar coordinate and one new base-4 digit in the height.

The proof has three parts:

1. prove the same-terminal-state pair law for H ;
2. use matching planar and height tags to extend it to every pair; and
3. apply the all-pairs law to a hypothetical collinear triple, then bound the successive tagged steps.

Section 2 supplies the Hilbert decoder and its finite-state properties. Section 3 proves the same-state pair law. Section 4 constructs the tagged walk and proves its all-pairs law. Sections 5 and 6 finish the theorem. Section 7 records verification and evidence, not an additional premise of the proof.

2 A nested discrete Hilbert path

2.1 A four-state decoder

On a bit pair $(x, y) \in \{0, 1\}^2$, define the four square transformations

$$\begin{aligned} I(x, y) &= (x, y), & S(x, y) &= (y, x), \\ T(x, y) &= (1 - y, 1 - x), & C(x, y) &= (1 - x, 1 - y). \end{aligned}$$

The transformations S and T are commuting involutions, with $S \circ T = T \circ S = C$. Here $f \circ g$ denotes function composition: $(f \circ g)(z) = f(g(z))$, so the right-hand transformation acts first. Thus

$$K = \{I, S, T, C\} \cong \mathbb{F}_2^2$$

is closed under composition. These are exactly the states reached by the decoder below.

Let

$$c_0 = (0, 0), \quad c_1 = (0, 1), \quad c_2 = (1, 1), \quad c_3 = (1, 0)$$

be the four child positions in the basic U-shaped traversal, and set

$$r_0 = S, \quad r_1 = I, \quad r_2 = I, \quad r_3 = T.$$

Here $r_q \in K$ is the digit-indexed transition map selected by digit q ; it is not a numeric sequence. For a length- ℓ base-4 word $w = q_{\ell-1} \cdots q_1 q_0$, read the digits from most significant to least significant. The mutable current state $g \in K$ starts at $g = I$. At digit q_j , emit one pair of coordinate bits and then replace the current state:

$$(x_j, y_j) = g(c_{q_j}), \quad g \leftarrow g \circ r_{q_j}.$$

The emitted pair at digit position j is used at binary place j :

$$H_\ell(w) = \left(\sum_{j=0}^{\ell-1} x_j 2^j, \sum_{j=0}^{\ell-1} y_j 2^j \right). \tag{1}$$

The reading direction and the place values point in opposite directions: the decoder processes $q_{\ell-1}, q_{\ell-2}, \dots, q_0$ in that order, while q_j writes the coordinate bits at binary place j . Accordingly, for two words “the first mismatch from the low end” means the least $j \geq 0$ for which their digits q_j differ. After the final digit q_0 , define $\sigma(w)$ to be the resulting value of g . Thus g denotes the changing intermediate state, while $\sigma(w)$ is its terminal value. This value is the accumulated quadrant transformation, not a point or an additional coordinate.

2.2 Hilbert properties and terminal states

Lemma 2.1 (Finite-order Hilbert properties). *For every $\ell \geq 0$, the map from length- ℓ base-4 words to $\{0, \dots, 2^\ell - 1\}^2$ given by (1) is a bijection. Words representing consecutive integers map to lattice neighbors. Moreover, for every length- ℓ word w ,*

$$H_{\ell+2}(00w) = H_\ell(w) \quad \text{and} \quad \sigma(00w) = \sigma(w).$$

Proof. First observe that the decoder respects its current state. Suppose a suffix is decoded from state g instead of I . If the I -started state at some later digit is h , the other state is $g \circ h$: both update on the right by the same r_q . The emitted pair is correspondingly $(g \circ h)(c_q) = g(h(c_q))$. Thus the entire suffix route decoded from g is the bitwise g -transform of the route decoded from I .

Induct on ℓ , recording simultaneously that the order- ℓ route starts at $(0, 0)$ and ends at $(2^\ell - 1, 0)$. The claim is immediate for $\ell = 0$. Put $N = 2^\ell$. At order $\ell + 1$, the first digit places four transformed copies of the order- ℓ route in four disjoint N -by- N quadrants. Bitwise S sends (x, y) to (y, x) , while bitwise T sends it to $(N - 1 - y, N - 1 - x)$. The four copies therefore have start-end pairs

$$\begin{aligned} (0, 0) &\longrightarrow (0, N - 1), \\ (0, N) &\longrightarrow (N - 1, N), \\ (N, N) &\longrightarrow (2N - 1, N), \\ (2N - 1, N - 1) &\longrightarrow (2N - 1, 0). \end{aligned}$$

The maps S and T only swap or reflect square coordinates, so they preserve unit ℓ^1 -adjacency inside each transformed copy. The three junction displacements are $(0, 1)$, $(1, 0)$, and $(0, -1)$; they give adjacency between copies. The four disjoint quadrants give bijectivity, and the displayed endpoints complete the simultaneous induction.

Finally, the first leading zero emits $(0, 0)$ and changes the state from I to S . The second emits $S(c_0) = (0, 0)$ and changes the state back to I . The remaining word is therefore decoded in its original state and at its original binary places, proving both padding identities. $\square$

For $n \in \mathbb{N}$, write n in base 4 and prepend zero digits to any sufficiently large even length. Lemma 2.1 makes both parts of the following definition independent of the chosen length.

Definition 2.2. The nested discrete Hilbert path $H: \mathbb{N} \rightarrow \mathbb{N}^2$ sends n to the finite-order point of its even-padded base-4 word. Its terminal state $\sigma(n) \in K$ is the transformation left by the same decoding.

To see the infinite claims explicitly, take $m \neq n$ and choose one even length large enough for both padded base-4 words. The words are distinct, so finite-order bijectivity gives distinct points; the padding identity identifies those points with $H(m)$ and $H(n)$. Thus H is injective. Likewise, represent n and $n + 1$ at one common even length. Finite-order adjacency, including across a base-4 carry, gives

$$\|H(n + 1) - H(n)\|_1 = 1 \quad (n \in \mathbb{N}). \quad (2)$$

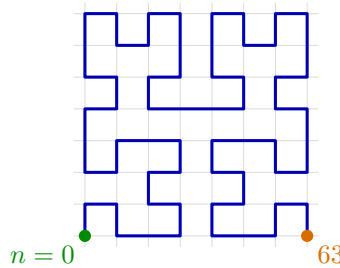


Figure 1: The order-three Hilbert route. Each segment is one planar lattice step; the same four-copy rule produces every higher order.

Only digits 0 and 3 alter the state. Since S and T commute and each undoes itself, for every finite word w ,

$$\sigma(w) = S^{N_0(w) \bmod 2} T^{N_3(w) \bmod 2}, \quad (3)$$

where $N_0(w)$ and $N_3(w)$ count its zero and three digits. Thus the terminal state remembers precisely the parities of those two counts.

For later use, every transition $g \mapsto g \circ r_q$ is invertible. If h is the outgoing state after digit q , then the incoming state was $h \circ r_q$, because r_q is an involution. Moreover $r_q(c_q) = c_q$, so the emitted pair can be recovered backwards as

$$(h \circ r_q)(c_q) = h(r_q(c_q)) = h(c_q). \quad (4)$$

This is the mechanism behind the pair law.

3 The two-adic pair law

For a nonzero planar chord u , remove its largest common power of two. The quantity $V(u)$ records twice that binary scale, plus one exactly when both remaining coordinates are odd. Equivalently, it distinguishes the two possible first appearances of the chord: one coordinate becomes nonzero at that scale, or both do.

Formally, put $\nu_2(0) = \infty$. For $u = (u_x, u_y) \in \mathbb{Z}^2 \setminus \{0\}$, define

$$p(u) = \min\{\nu_2(u_x), \nu_2(u_y)\}, \quad (5)$$

$$\epsilon(u) = \begin{cases} 1, & \nu_2(u_x) = \nu_2(u_y), \\ 0, & \nu_2(u_x) \neq \nu_2(u_y), \end{cases} \quad (6)$$

$$V(u) = 2p(u) + \epsilon(u). \quad (7)$$

Lemma 3.1 (Scaling). *For $u \neq 0$ and every positive integer k ,*

$$V(ku) = V(u) + 2\nu_2(k).$$

Proof. Multiplication by k adds $\nu_2(k)$ to each finite coordinate valuation. It therefore adds $\nu_2(k)$ to $p(u)$ and leaves the equality bit $\epsilon(u)$ unchanged. $\square$

The factor 2 has a concrete meaning. Multiplication by 2 shifts both coordinate bit streams one place, while V reserves the two consecutive values $2j$ and $2j + 1$ for the one-coordinate and two-coordinate appearances at scale j . A shift of one scale must therefore add two to V .

Lemma 3.2 (First mismatch). *Let $q_{\ell-1} \cdots q_0$ and $q'_{\ell-1} \cdots q'_0$ be equally long, even-padded base-4 words with the same terminal state, and let*

$$j = \min\{k \geq 0 : q_k \neq q'_k\}.$$

Thus positions $0, \dots, j-1$ form their common low suffix. Their Hilbert coordinates agree below binary place j . At place j , exactly one coordinate bit differs when $q_j - q'_j$ is odd, and both coordinate bits differ when $q_j - q'_j \equiv 2 \pmod{4}$.

Proof. The words have one common low-digit suffix. Start from their equal terminal states and undo that suffix in lockstep. Invertibility of every transition shows that immediately after the mismatching digit both decoders have one common state h .

Let the mismatching digits be $d \neq e$. Their incoming states were $h \circ r_d$ and $h \circ r_e$. By (4), the emitted pairs at the mismatch are $h(c_d)$ and $h(c_e)$. The corners c_0, c_1, c_2, c_3 form a cyclic Gray code: c_d and c_e differ in exactly one coordinate when $d - e$ is odd, and in both coordinates when $d - e \equiv 2 \pmod{4}$. Every $h \in K$ only swaps coordinates and complements coordinate bits. It therefore permutes which coordinates differ but preserves whether the number of differing coordinates is one or two. Starting from the common state h , the identical lower suffix also emits identical bits below place j . $\square$

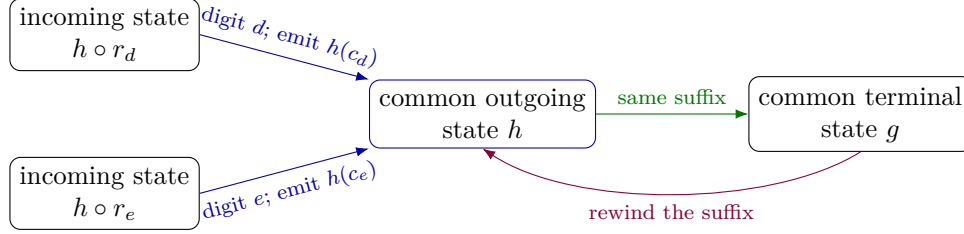


Figure 2: The first-mismatch alignment. Forward decoding sends the two possibly different incoming states through digits d and e to one common outgoing state h . Equal terminal states let the shared low suffix be rewound from g back to h ; rewinding changes state, not emitted bits.

Theorem 3.3 (Same-terminal-state pair law). *If $m \neq n$ and $\sigma(m) = \sigma(n)$, then*

$$V(H(n) - H(m)) = \nu_2(|n - m|). \quad (8)$$

Proof. By symmetry assume $m < n$. Pad both base-4 words to one common even length, writing the digits of m as $q_{\ell-1} \cdots q_0$ and those of n as $q'_{\ell-1} \cdots q'_0$. Let j be their first differing position from the low end, and put $d = q_j$ and $e = q'_j$. Then $d \neq e$ and

$$n - m = 4^j M, \quad M \equiv e - d \pmod{4}.$$

If $e - d$ is odd, then $\nu_2(n - m) = 2j$. By Lemma 3.2, exactly one coordinate of $H(n) - H(m)$ has differing bits at position j ; in that coordinate the difference has the form

$$\pm 2^j + 2^{j+1} L = 2^j (\pm 1 + 2L)$$

for some integer L . The bracket is odd, so higher binary places cannot cancel the mismatch or add another factor of two. The other coordinate has larger valuation. Thus $p = j$, $\epsilon = 0$, and $V = 2j$.

If $e - d \equiv 2 \pmod{4}$, then $\nu_2(n - m) = 2j + 1$. The same odd-bracket calculation applies to both coordinate differences, so both have valuation j . Hence $p = j$, $\epsilon = 1$, and $V = 2j + 1$. These cases exhaust unequal base-4 digits. $\square$

4 Tagging all four terminal states

The pair law compares indices with one common terminal state. We remove that restriction by recording the state in one additional planar bit pair and in the corresponding residue modulo four. Label the four states in cyclic order by

$$\lambda(I) = 0, \quad \lambda(S) = 1, \quad \lambda(C) = 2, \quad \lambda(T) = 3,$$

and give $g \in K$ the Gray-code corner tag

$$u_g = c_{\lambda(g)}.$$

Thus

g	I	S	C	T
$\lambda(g)$	0	1	2	3
u_g	(0, 0)	(0, 1)	(1, 1)	(1, 0).

Define

$$\begin{aligned} G(n) &= 2H(n) + u_{\sigma(n)} \in \mathbb{Z}^2, \\ z(n) &= 4n + \lambda(\sigma(n)) \in \mathbb{Z}, \\ P_n &= (G(n), z(n)) \in \mathbb{Z}^3. \end{aligned} \tag{9}$$

Multiplication by two leaves one low binary place for the planar corner tag, while multiplication by four leaves one low base-4 place for the height label.

Theorem 4.1 (All-pairs tagged law). *For all distinct $m, n \in \mathbb{N}$, the planar chord is nonzero and*

$$V(G(n) - G(m)) = \nu_2(|z(n) - z(m)|). \tag{10}$$

Proof. Put

$$d = n - m, \quad i = \lambda(\sigma(m)), \quad j = \lambda(\sigma(n)), \quad X = H(n) - H(m).$$

Then

$$G(n) - G(m) = 2X + c_j - c_i, \quad z(n) - z(m) = 4d + j - i. \tag{11}$$

If $i = j$, the states agree and the corner tags cancel. Theorem 3.3 and Lemma 3.1 give

$$V(2X) = V(X) + 2 = \nu_2(|d|) + 2 = \nu_2(|4d|).$$

Suppose $i \neq j$. If $j - i$ is odd, then c_i and c_j are adjacent corners in the cyclic Gray code. Exactly one coordinate of $2X + c_j - c_i$ is odd, so this chord is nonzero and has $V = 0$. The integer $4d + j - i$ is also odd.

The remaining case is $j - i = \pm 2$. The two corners are opposite, so both coordinates of $2X + c_j - c_i$ are odd. Hence $p = 0$, $\epsilon = 1$, and $V = 1$. Also

$$4d + j - i = 4d \pm 2 = 2(2d \pm 1)$$

has valuation one. These cases prove (10). $\square$

5 Excluding collinear triples

Theorem 5.1 (The tagged lift is triple-free). *The sequence $(P_n)_{n \geq 0}$ in (9) contains no three collinear terms.*

Proof. Because λ takes values in $\{0, 1, 2, 3\}$,

$$1 \leq z(n+1) - z(n) = 4 + \lambda(\sigma(n+1)) - \lambda(\sigma(n)) \leq 7. \tag{12}$$

Thus z is strictly increasing. Suppose for contradiction that $a < b < c$ and P_a, P_b, P_c are collinear. Write the adjacent height gaps and planar chords as

$$\begin{aligned} A &= z(b) - z(a), & B &= z(c) - z(b), \\ X &= G(b) - G(a), & Y &= G(c) - G(b). \end{aligned}$$

Both A and B are positive. Collinearity makes the full displacements (X, A) and (Y, B) proportional, and their last coordinates force

$$BX = AY. \tag{13}$$

Write

$$A = gr, \quad B = gs, \quad g = \gcd(A, B), \quad \gcd(r, s) = 1.$$

After cancelling g , equation (13) becomes $sX = rY$. Coordinatewise coprimality gives an integer vector W such that

$$X = rW, \quad Y = sW. \tag{14}$$

Theorem 4.1 gives $X \neq 0$, hence $W \neq 0$.

Apply the all-pairs law and scaling lemma to the two adjacent pairs:

$$\begin{aligned} \nu_2(g) + \nu_2(r) &= V(X) = V(W) + 2\nu_2(r), \\ \nu_2(g) + \nu_2(s) &= V(Y) = V(W) + 2\nu_2(s). \end{aligned}$$

Therefore $\nu_2(r) = \nu_2(s)$. Coprimality forces both valuations to be zero, so r and s are odd, and either equality gives $V(W) = \nu_2(g)$.

For the endpoint pair, the height gap is $A+B = g(r+s)$ and the planar chord is $X+Y = (r+s)W$. If $t = \nu_2(r+s)$, the all-pairs law gives

$$\nu_2(g) + t = V((r+s)W) = V(W) + 2t = \nu_2(g) + 2t.$$

Thus $t = 0$, so $r+s$ is odd. But r and s are both odd, making their sum even. This contradiction proves the theorem. $\square$

6 The finite step menu

For one successive step, (9) gives

$$\begin{aligned} G(n+1) - G(n) &= 2(H(n+1) - H(n)) + u_{\sigma(n+1)} - u_{\sigma(n)}, \\ z(n+1) - z(n) &= 4 + \lambda(\sigma(n+1)) - \lambda(\sigma(n)). \end{aligned}$$

The first planar term is twice a coordinate unit vector by (2); each coordinate of the tag difference lies between -1 and 1 . Equation (12) bounds the last coordinate. Every successive displacement therefore belongs to the fixed finite set

$$S = \{(d_x, d_y, d_z) \in \mathbb{Z}^3 : |d_x| \leq 3, |d_y| \leq 3, 1 \leq d_z \leq 7\}. \tag{15}$$

The strictly increasing height makes (P_n) infinite with no repeated terms, and Theorem 5.1 excludes collinear triples. This proves Theorem 1.1.

7 Verification and evidence

Proof boundary. The proof of Theorem 1.1 ends above. Nothing in this section is an additional premise; it records formal verification and the status of external claims.

7.1 Lean formalization

The repository [6] contains a Lean 4 formalization in `formal/Hilbert193`. It uses Lean 4.33.0 and the corresponding pinned Mathlib revision. The final theorem is `Hilbert193.erdos193_unconditional` in `Hilbert193/Continuity.lean`. It constructs:

1. a finite set of integer three-dimensional step vectors;
2. an explicit sequence $\mathbb{N} \rightarrow \mathbb{Z}^3$ whose successive displacements lie in that set; and
3. a proof that every ordered triple of distinct indices fails the exact integer collinearity equations.

The Lean predicate packages collinearity as the two integer cross-multiplication equalities forced by the strictly increasing third coordinate. As shown in the proof above, this is the denominator-free form of ordinary real affine collinearity for the ordered points in this walk.

The axiom audit in `Hilbert193/AxiomAudit.lean` reports only `propext`, `Classical.choice`, and `Quot.sound`, the standard axioms used by Mathlib. No theorem in the construction depends on a finite computation. From the repository root, the formal proof is checked by

```
cd formal/Hilbert193
lake build
lake env lean Hilbert193/AxiomAudit.lean
```

7.2 Finite implementation check

The standalone script `viz/hilbert_walk_tagged_demo.py` independently evaluates the decoder, constructs a finite tagged prefix, and checks the all-pairs identity, the step bounds, and every collinear triple in that prefix using exact integer arithmetic. This is an implementation check only; no finite computation proves the infinite theorem.

7.3 Logical and historical status

The theorem above is unconditional. The Lean development checks the complete tagged construction, while the standalone script checks an implementation and a finite prefix only. Novelty, priority, and correspondence with the intended reading of Erdős Problem 193 remain subject to external mathematical review and community acceptance.

Acknowledgments and disclosure

Stijn Cambie suggested the state-tagging refinement that removes the same-terminal-state selector and shared a GPT-assisted alternative draft. The author thanks him for this substantial simplification.

The working stack included Pi (<https://pi.dev/>), Oh My Pi (<https://omp.sh/>), Paseo (<https://paseo.sh/>), and OpenAI GPT-5.6 for proof exploration, Lean code drafting, computational verification tooling, visualization, and manuscript preparation. The author directed the work, checked the generated artifacts, and takes responsibility for every claim. The proof is kernel-checked, but at the date of this preprint it has not yet completed external mathematical peer review.

References

- [1] T. F. Bloom, Erdős Problem #193, <https://www.erdosproblems.com/193>, accessed August 19, 2026.

- [2] J. L. Gerver and L. T. Ramsey, On certain sequences of lattice points, *Pacific Journal of Mathematics* **83** (1979), no. 2, 357–363. <https://doi.org/10.2140/pjm.1979.83.357>.
- [3] D. Hilbert, Über die stetige Abbildung einer Linie auf ein Flächenstück, *Mathematische Annalen* **38** (1891), 459–460.
- [4] L. de Moura and S. Ullrich, The Lean 4 theorem prover and programming language, in *Automated Deduction—CADE 28*, Lecture Notes in Computer Science 12699, Springer, 2021, 625–635.
- [5] The Mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, 2020, 367–381.
- [6] E. Kalviainen, Erdős 193: state-tagged Hilbert lift, source code and formal proof, <https://github.com/ekalvi/erdos-193>, 2026.
- [7] H. Sagan, *Space-Filling Curves*, Springer, 1994.